

Mobile & Application Development (Android)

(CSI-601)

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Why Android



Android Features...

Sr.	Feature & Description
1	<i>Beautiful UI</i> Android OS basic screen provides a beautiful and intuitive user interface.
2	<i>Connectivity</i> Bluetooth, Wi-Fi
3	<i><u>Storage</u></i> <u>SQLite</u> , a lightweight relational database, for data storage purposes.
4	<i>Media Support</i> H.263, H.264, MPEG-4 SP, AMR, AMR-WB, AAC, HE-AAC, AAC 5.1, MP3, MIDI, Ogg Vorbis, WAV, JPEG, PNG, GIF, and BMP.
5	<i>Messaging</i> SMS and MMS
6	<i>Web browser</i> Based on the open-source <u>WebKit layout engine</u> , coupled with <u>Chrome's V8 JavaScript engine</u> supporting HTML5 and CSS3.

Android Features...

7	<i>Multi-touch</i> Android has native support for multi-touch which was initially made available in handsets such as the HTC Hero.
8	<i>Multi-tasking</i> User can jump from one task to another and same time various application can run simultaneously.
9	<i>Resizable widgets</i> Widgets are resizable, so users can expand them to show more content or shrink them to save space.
10	<i>Multi-Language</i> Supports.
11	<i>GCM</i> <u>Google Cloud Messaging (GCM)</u> is a service that lets developers send short message data to their users on Android devices, without needing a proprietary sync solution. (<i>Direct Delivery/Communication</i>)
12	<i>Wi-Fi Direct</i> A technology that lets apps discover and pair directly, over a high-bandwidth
13	<i>Android Beam</i> A popular NFC-based technology that lets users instantly share, just by touching two NFC-enabled phones together.

Android Applications

- Android applications are usually developed in the Java language using the Android Software Development Kit.
- Layouts (Screens / Static Pages) are designed in XML format and are programmatically implemented by the Java coding. (Application isolation form code)
- Once developed, Android applications can be packaged easily and sold out either through a store such as **Google Play, SlideME, Opera Mobile Store, Mobango, F-droid** and the **Amazon Appstore**.
- Android powers hundreds of millions of mobile devices in more than 190 countries around the world. It's the largest installed base of any mobile platform and growing fast. Every day more than 1 million new Android devices are activated worldwide.

Categories of Android Applications



Music



News



Multimedia



Sports



Lifestyle



Food & Drink



Travel



Weather



Books



Business



Reference



Navigation



Social Media



Utilities



Finance

Android OS

Android is a mobile operating system based on a modified version of the Linux kernel and other open-source software, designed primarily for touch screen mobile devices such as smart phones and tablets.

Initially Developed by: Andy Rubin , Nick Sears & Rich Miner (2003)

Current Developer: Google Inc. (since 2005)

Programming languages: [Java](#), [Kotlin](#), [JavaScript](#), [C](#), [Go](#), [C++](#), [Assembly language](#)

Default user interface: [Graphical user interface](#) ([Multi-touch](#))

Android OS Versions

- The Organization was established in the year 2003 and later owned by the Google in 2005.
- The first android beta version was released in 2007.
- The initial releases were named alphabetically from A to P. Then version 10 was release leading to the current latest release which is **Android version 15**. (released in year 2024), Latest Version is 16 development

